HW9-Prometheus & Grafana

Packer build & Update new AMI ID

```
yj@YJs-MacBook-Air hw8 % cd packer
yj@YJs-MacBook-Air packer % ll
.rw-r--r-- yj staff 2.1 KB Tue Mar 31 19:53:33 2026 ami.pkr.hcl
yj@YJs-MacBook-Air packer % packer init .
yj@YJs-MacBook-Air packer % packer build ami.pkr.hcl
amazon-ebs.amazon-linux: output will be in this color.

==> amazon-ebs.amazon-linux: Prevalidating any provided VPC information
==> amazon-ebs.amazon-linux: Prevalidating AMI Name: custom-docker-monitoring-ami-1775012833
==> amazon-ebs.amazon-linux: Found Image ID: ami-014d82945a82dfba3
==> amazon-ebs.amazon-linux: Creating temporary keypair: packer_69cc8be1-d948-c63c-645d-66f3d2974b37
==> amazon-ebs.amazon-linux: Creating temporary security group for this instance: packer_69cc8be3-fe56-6c4b-02e1-957d00efceee
==> amazon-ebs.amazon-linux: Authorizing access to port 22 from [0.0.0.0/0] in the temporary security groups...
==> amazon-ebs.amazon-linux: Launching a source AWS instance...
==> amazon-ebs.amazon-linux: Instance ID: i-053eade755f2e6dfd
==> amazon-ebs.amazon-linux: Waiting for instance (i-053eade755f2e6dfd) to become ready...
```

```
==> amazon-ebs.amazon-linux: Provisioning with shell script: /var/folders/km/qcwvmr4x5j14x85706hdvb3w0000gn/T/packer-shell1629836300
==> amazon-ebs.amazon-linux: Stopping the source instance...
==> amazon-ebs.amazon-linux: Stopping instance
==> amazon-ebs.amazon-linux: Waiting for the instance to stop...
==> amazon-ebs.amazon-linux: Creating AMI custom-docker-monitoring-ami-1775012833 from instance i-053eade755f2e6dfd
==> amazon-ebs.amazon-linux: Attaching run tags to AMI...
==> amazon-ebs.amazon-linux: AMI: ami-05c5ce15913ecb0b3
==> amazon-ebs.amazon-linux: Waiting for AMI to become ready...
==> amazon-ebs.amazon-linux: Skipping Enable AMI deprecation...
==> amazon-ebs.amazon-linux: Skipping Enable AMI deregistration protection...
==> amazon-ebs.amazon-linux: Terminating the source AWS instance...
==> amazon-ebs.amazon-linux: Cleaning up any extra volumes...
==> amazon-ebs.amazon-linux: No volumes to clean up, skipping
==> amazon-ebs.amazon-linux: Deleting temporary security group...
==> amazon-ebs.amazon-linux: Deleting temporary keypair...
Build 'amazon-ebs.amazon-linux' finished after 4 minutes 18 seconds.

==> Wait completed after 4 minutes 18 seconds

==> Builds finished. The artifacts of successful builds are:
--> amazon-ebs.amazon-linux: AMIs were created:
us-west-2: ami-05c5ce15913ecb0b3
```

Terraform Init

```
yj@YJs-MacBook-Air terraform % terraform init
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.38.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
yj@YJs-MacBook-Air terraform %
```

Terraform Apply

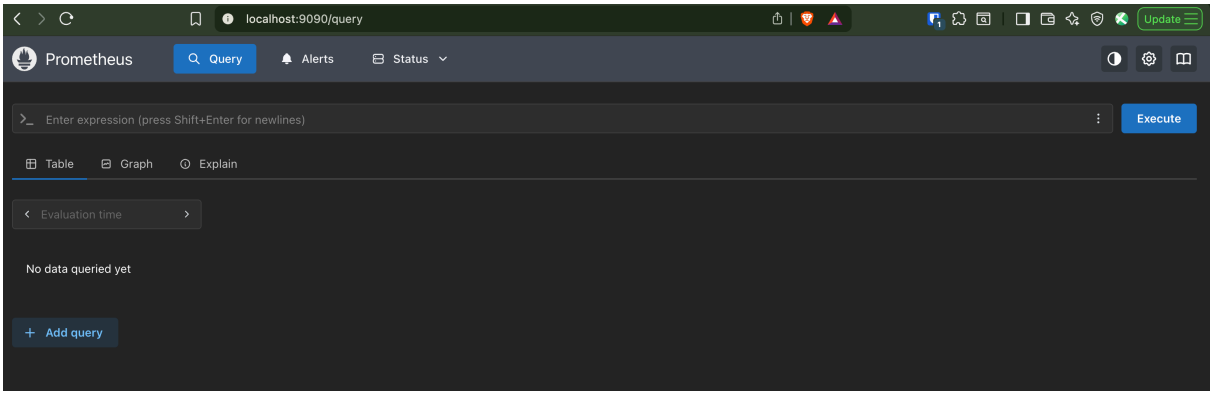
```
yj@YJs-MacBook-Air terraform % terraform apply
data.aws_ami.amazon_linux: Reading...
data.aws_ami.amazon_linux: Read complete after 1s [id=ami-014d82945a82dfba3]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.bastion will be created
  + resource "aws_instance" "bastion" {
      + ami                                  = "ami-014d82945a82dfba3"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = true
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "hw8-deployer-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
```

```
Apply complete! Resources: 26 added, 0 changed, 0 destroyed.

Outputs:

bastion_public_ip = "34.223.7.150"
monitoring_private_ip = "10.0.3.137"
private_instance_ips = [
  "10.0.3.126",
  "10.0.4.117",
  "10.0.3.163",
  "10.0.4.247",
  "10.0.3.219",
  "10.0.4.182",
]
yj@YJs-MacBook-Air terraform %
```

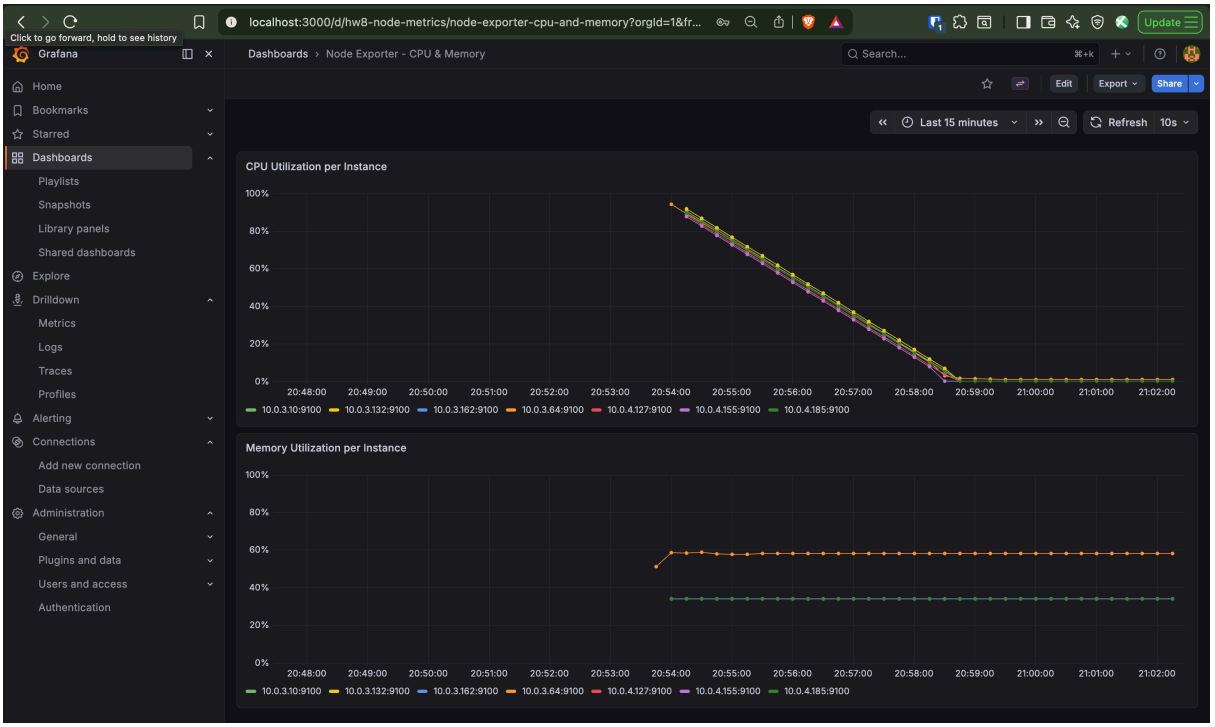
Access via ssh port mapping

```
yj@YJs-MacBook-Air terraform % ssh -A -L 9090:10.0.3.137:9090 -L 3000:10.0.3.137:3000 ec2-user@34.223.7.150
The authenticity of host '34.223.7.150 (34.223.7.150)' can't be established.
ED25519 key fingerprint is: SHA256:SKZrQ2Shd2OYFPh3mjWGR0JJQJY2/I3y44OOfF6iOPs
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.223.7.150' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-1-135 ~]$
```

Prometheus



Grafana



Destroy

```
aws_security_group.private: Destroying... [id=sg-0b449ffef3d63ea7b]
aws_key_pair.deployer: Destruction complete after 1s
module.vpc.aws_subnet.private[0]: Destruction complete after 1s
module.vpc.aws_subnet.private[1]: Destruction complete after 1s
aws_security_group.private: Destruction complete after 1s
aws_security_group.monitoring: Destroying... [id=sg-0914b4f0413c51326]
module.vpc.aws_internet_gateway.igw: Still destroying... [id=igw-068e85bd7341ce200, 01m00s elapsed]
module.vpc.aws_nat_gateway.nat: Still destroying... [id=nat-057bf9b2087624870, 01m00s elapsed]
aws_security_group.monitoring: Destruction complete after 1s
aws_security_group.bastion: Destroying... [id=sg-0f84cf4ef29f9dbba]
aws_security_group.bastion: Destruction complete after 0s
module.vpc.aws_nat_gateway.nat: Still destroying... [id=nat-057bf9b2087624870, 01m10s elapsed]
module.vpc.aws_internet_gateway.igw: Still destroying... [id=igw-068e85bd7341ce200, 01m10s elapsed]
module.vpc.aws_nat_gateway.nat: Destruction complete after 1m11s
module.vpc.aws_eip.nat: Destroying... [id=eipalloc-0562dc2cbe8f5cb01]
module.vpc.aws_subnet.public[0]: Destroying... [id=subnet-074aabdeece97a0b3]
module.vpc.aws_subnet.public[1]: Destroying... [id=subnet-0f01f76341a95a079]
module.vpc.aws_subnet.public[0]: Destruction complete after 1s
module.vpc.aws_subnet.public[1]: Destruction complete after 1s
module.vpc.aws_eip.nat: Destruction complete after 1s
module.vpc.aws_internet_gateway.igw: Destruction complete after 1m19s
module.vpc.aws_vpc.main: Destroying... [id=vpc-0aac2bd160984318a]
module.vpc.aws_vpc.main: Destruction complete after 1s

Destroy complete! Resources: 26 destroyed.
```